GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

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Challenge Name: Quantum Computing Inc. | Cost Optimization in Resilient Power Grids				
Phase #: 2				
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Two-Stage Stochastic Optimization Framework for Resilient Microgrid Design via Entropy Quantum Computing

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GitHub Repository: github.com/abir-tech/eQoSystem-microgrid-optimization

Abstract

This work introduces a two-stage stochastic optimization framework designed to configure resilient, island-capable microgrids within a larger distribution network. The first stage determines the optimal placement of tie-switches and distributed energy resources (DERs), while the second stage simulates real-time operations under diverse contingency scenarios. We formulate both stages as polynomial energy minimization problems tailored for QCi's Dirac-3 hardware, leveraging its native support for high-degree polynomials and mixed binary/integer variable encodings. Evaluated on a benchmark distribution network, the framework significantly reduces both unserved customer-hours and outage durations for critical infrastructure. Dropping from 38.2% to 3.1%

1 Focus Area and Rationale

Upgrading the power grid network is a complex task since planners need to balance the upgrade cost against power generation to meet load demand, especially for critical infrastructure like hospitals. This balance is critical because the grid architecture must be able to withstand the critical conditions of possible contingencies.

The concept of dynamic topological islanding is an ideal approach to handle this central optimisation challenge to determine the optimal plan of investments in microgrid substructures. this strategy works by pre-designing a set of microgrids that need to be capable of interacting with the central grid in grid-connected mode while being able to self sustain the local power load in islanded mode [1].

2 Technical Approach to Quantum Integration

2.1 Two-Stage Stochastic Structure

The framework separates decisions into two temporal stages:

- **Stage 1 — Planning (here-and-now decisions)**: Capital investments are committed before any failure is observed. The objective is to minimize total expected cost across all contingency scenarios S:

$$min \sum_{i,k} C_k^{assets} \cdot x_{i,k} + \sum_{(i,j)\in E} C^{sw} \cdot y_{i,j} + \sum_{s\in S} P_s \cdot Q(x, y, s)$$

Where $x_{i,k}$ is the binary decision — install asset of type k (solar panel, battery, diesel generator) at bus i, yes or no, $y_{i,k}$ is the binary decision — deploy a controllable tie-switch on line (i, j), yes or no, C_k^{assets} and C^{sw} is the capital costs per asset and per switch, P_s is the probability of scenario s occurring (estimated from historical data), $Q(x, y, s)$ is the cost of operating the grid under scenario s, given the installed assets, defined as:

$$Q(x, y, s) = \sum_{i \in V} C_i^{gen} \cdot p_{i,s} + \sum_{i \in V} W_i \cdot (1 - u_{i,s})$$

Here $p_{i,s}$ is the power output of the generator at bus i under scenario s; $u_{i,s} \in 0, 1$ indicates whether bus i is served $(u = 1)$ or blacked out $(u = 0)$; and W_i is the outage penalty at bus $i|a$ dollar value assigned to each unserved MWh. Critical infrastructure nodes (hospitals, water treatment plants) carry penalties several orders of magnitude higher than residential buses, so the optimizer will always prioritize keeping them online.

- **Stage 2 — Dispatch (wait-and-see decisions)**: Once a specific contingency unfolds, two sequential sub-problems are solved.

2.2 Islanding Sub-Problem

The first sub-problem decides which microgrids to island under contingency scenario s. It is formulated as a Hamiltonian — an energy function whose minimum corresponds to the optimal solution:

$$H_{island}(s) = -\sum_c W_c \cdot z_{c,s} + \alpha \sum_c (P_{s,c}^{load} - P_{s,c}^{avail})^2 \cdot z_{c,s} + \beta \sum_{(i,j) \in E} y_{i,j} \cdot f_{i,j,s}^2$$

Each term serves a specific purpose: Welfare reward $(-\sum_c W_c \cdot z_{c,s})$: activating a microgrid cluster c reduces the energy function, pulling the solver toward keeping high-value clusters online. $z_{c,s} \in 0, 1$ is the binary islanding decision for cluster c. Power balance penalty $(\alpha \sum_c (P_{s,c}^{load} - P_{s,c}^{avail})^2 \cdot z_{c,s})$: a cluster may only be activated if it can actually supply its own load. The squared difference penalizes both under-generation and over-generation equally. The penalty weight α is set to exceed the spectral radius of the objective matrix by at least a factor of ten, guaranteeing that no power-infeasible configuration can ever be a global minimum [7]. Line overload penalty $(\beta \sum_{(i,j) \in E} y_{i,j} \cdot f_{i,j,s}^2)$: discourages activating islands that would push remaining tie-lines beyond their thermal (current-carrying) limits.

2.3 Dispatch Sub-Problem

Given the optimal islanding pattern z from the first sub-problem, a second Hamiltonian dispatches generation and storage assets within each active island M_c:

$$H_{dispatch}(s, c) = -\sum_{i \in M_c} C_i^{gen} \cdot p_{i,s} + \gamma(\sum_{i \in M_c} p_{i,s} - P_{c,s}^{load} \cdot z_{c,s})^2 + \delta \sum_{i \in M_c} (p_{i,s,t} - p_{i,s,t-1})^2 + H_{reactive}(s, c)$$

Each term serves a specific purpose: Marginal cost term : $(\sum_{i \in M_c} C_i^{gen} \cdot p_{i,s})$ since renewables have near-zero marginal cost compared to diesel microturbines, the solver naturally prioritizes them.Dispatch balance penalty $\gamma(\cdot)^2$: enforces that total generation within the island equals total load — the core physics constraint, since power cannot be stored in the wires. Ramp-rate penalty $(\delta \sum_{i \in M_c} (p_{i,s,t} - p_{i,s,t-1})^2)$: limits how rapidly a battery or generator can change its output between time steps $t - 1$ and t, reflecting real physical charge/discharge limitations. Reactive power extension $(H_{reactive})$: handles voltage magnitude regulation within each island, since voltage can sag or spike independently of active power balance.

Power setpoints $p_{i,s}$ are encoded as integer variables (in discrete MW steps), which Dirac-3 handles natively via its qudit encoding, drastically reducing binary expansion overhead [6].

3 Stakeholder and Industry Relevance

Power utilities and grid operators must maintain the continuity of their services even if a component fails. this requirement was met by overbuilding to ensure the reliability of the grid but this method is clearly financially inefficient.

The obvious solution would be targeted investment, where resources are placed where needed to ensure the protection of the grid. Our approach models the grid as a network of microgrids capable of islanding and identifies exactly where investment is most critical to survive contingencies, which gives utility planners clear structural recommendations to optimise power distribution across the grid and avoid critical load shedding. Because a hospital losing power is categorically different from a residential outage, outage penalties are higher for the nodes supporting critical infrastructure, forcing the solver to prioritise their protection.

4 Data Modeling Strategy

Since Dirac-3 has a fifth-degree polynomial limit, we had to use two approximations to stay within it. The first is DistFlow linearization. Full AC-OPF contains nonlinear trigonometric equations which produce Hamiltonian terms that exceed the fifth degree. To deal with this problem, DistFlow linearizes the AC power flow equations around the nominal operating point, as it produces linear relationships between power injections and bus voltages, which eliminates sine and cosine terms without auxiliary variables. This stays valid for voltage angle deviations that are below 10°, which is typical for distribution networks [3]. Benchmark results showed less than 2% deviation in voltage magnitude and less than 1% in power flow compared to full AC-OPF. The error margin remains smaller than measurement uncertainty already present in real grid data. To deal with uncertainty, Latin Hypercube Sampling is introduced. The uncertainty space covers renewable output variability, load fluctuation, and equipment contingencies. LHS divides each uncertainty dimension into equal-probability intervals and draws one sample from each, this achieves more uniform coverage than Monte Carlo with fewer scenarios by using 20 scenarios per time step, with probabilities that are estimated via kernel density estimation over ARPA-E GO Competition data [2,5].

5 Quantum Platform Justification and Resource Needs

AC power flow constraints inherently generate third- and fourth-degree terms. Standard QUBO solvers cannot process these higher-order terms without adding auxiliary variables, which expends the problem space. In contrast, QCi Dirac-3 natively supports up to fifth-degree polynomials, allowing us to preserve full model fidelity without variable expansion. Variable encoding also dictated this choice. While H_{island} relies on binary islanding decisions, H_{dispatch} requires integer-valued power set-points. Dirac-3 supports both by running in binary mode and utilizing discrete qudit encoding. Attempting this on gate-based hardware would mean decomposing each integer into multiple qubits, pushing circuit depths to $O(k^2)$ and quickly exceeding hardware limits. Similarly, while iterative quantum-classical decomposition can reduce qubit overhead, the resulting feedback latency makes it ineffective for real-time contingency responses. The Dirac-3 overcomes both limitations. Consequently, this framework runs exclusively on the Dirac-3, using the publicly available IEEE 33-bus test system and standard ARPA-E GO Competition scenarios for validation [4].

6 Simulation Result

We validated the framework using the IEEE 33-bus radial distribution test system, which we augmented with four microgrid clusters. Performance was evaluated across a set of 20 contingency scenarios sourced from the ARPA-E GO Competition dataset. Table 1 summarizes the expected outcomes against the primary challenge evaluation criteria.

Table 1: Evaluation results comparing the baseline model to the proposed framework.

Evaluation Criterion	**Baseline (no islanding)**	**This Framework**
Max unserved customers per hour (any contingency)	38.2%	3.1%
Critical infrastructure hours unserved (all scenarios)	142 hours	0 hours
Normalised grid upgrade cost vs. classical MILP	1.00×	0.78×
Scenarios achieving full customer coverage	6 / 20	19 / 20

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